

Week 17: Security & Vulnerability Scanning

Date: June 29, 2026

Study Time: 5-8 hours total

GOAL

Identify and remediate security issues in your network

STUDY TASKS (1-2 hours)

Reading

☐ "How Linux Works" - Chapter 17 (Security Best Practices)

Video Learning

☐ LearnLinuxTV - "Nmap Security Scanning"

☐ Professor Messer - "Vulnerability Assessment"

Concepts

☐ Port scanning basics

☐ Service fingerprinting

☐ CVE (Common Vulnerabilities and Exposures)

☐ CVSS (Common Vulnerability Scoring System)

☐ Security hardening

☐ Attack surface reduction

☐ Defense in depth

LAB WORK (3-4 hours)

Install Security Tools

☐ Install nmap (network scanner)

```
bash
```

```
# Ubuntu:
sudo apt install nmap

# Fedora:
sudo dnf install nmap

# Verify
nmap --version
```

☐ Install Lynis (security auditing)

```
bash
```

```
# Ubuntu:
sudo apt install lynis

# Fedora:
sudo dnf install lynis

# Verify
lynis --version
```

Port Scanning with Nmap

 **IMPORTANT:** Only scan YOUR OWN devices! Scanning others' networks may be illegal.

☐ Basic network scan

```
bash
```

```
# Scan your Tailscale network
nmap 100.64.0.0/10

# Scan specific IP
nmap 100.64.0.10 # Your Fedora server
```

☐ Scan with service detection

```
bash
```

```
nmap -sV 100.64.0.10

# -sV = Version detection
# Shows what services are running and their versions
```

☐ Scan specific ports

bash

```
# Scan common ports
nmap -p 22,80,443,3000 100.64.0.10

# Scan port range
nmap -p 1-1000 100.64.0.10

# Scan ALL ports (1-65535) - slow!
nmap -p- 100.64.0.10
```

☐ Fast scan (top 100 ports)

bash

```
nmap -F 100.64.0.10
```

☐ Aggressive scan (service + OS detection)

bash

```
nmap -A 100.64.0.10

# Includes:
# - Version detection
# - OS detection
# - Script scanning
# - Traceroute
```

☐ Output to file

bash

```
nmap -oN scan-results.txt 100.64.0.10
# -oN = normal output

# Or all formats:
nmap -oA scan-all-formats 100.64.0.10
# Creates: .nmap, .xml, .gnmap
```

Scan All Your Devices

☐ Create comprehensive scan script

bash

```
nano ~/scan-infrastructure.sh

#!/bin/bash

SCAN_DATE=$(date +%Y%m%d_%H%M%S)
REPORT_DIR="$HOME/security-scans"

# Create report directory
mkdir -p "$REPORT_DIR"

# Array of your devices
declare -a devices=(
    "100.64.0.10:Fedora_Server"
    "100.64.0.20:Pi_Zero"
    "100.64.0.30:Pi_4"
)

echo "=== Infrastructure Security Scan ==="
echo "Started: $(date)"
echo ""

for device in "${devices[@]"; do
    IP="${device%%:*}"
    NAME="${device##*:}"

    echo "Scanning: $NAME ($IP)"

    # Quick scan
    nmap -sV -p- "$IP" -oN "$REPORT_DIR/${NAME}_${SCAN_DATE}.txt"

    if [ $? -eq 0 ]; then
        echo "  ✓ Scan complete"
    else
        echo "  × Scan failed"
    fi
done

echo ""
echo "=== Scan Complete ==="
echo "Reports saved to: $REPORT_DIR"
echo "Finished: $(date)"

chmod +x ~/scan-infrastructure.sh
```

☐ Run the scan

```
bash

./scan-infrastructure.sh

# Review results
ls -lh ~/security-scans/
cat ~/security-scans/Fedora_Server_*.txt
```

Security Auditing with Lynis

Lynis performs comprehensive security audits.

☐ Run system audit on laptop

```
bash

# Run audit
sudo lynis audit system

# Takes 2-5 minutes
# Watch for warnings and suggestions
```

☐ Review Lynis report

```
bash

# Report location:
cat /var/log/lynis.log

# Or in some systems:
cat /var/log/lynis-report.dat
```

☐ Key sections to review

Look for:

- Warnings (things that need attention)
- Suggestions (improvements)
- Hardening index (percentage)

Common findings:

- Unneeded services running
- Weak file permissions
- Missing security updates
- Unnecessary packages installed
- Kernel parameters not optimized

Run Lynis on All Devices

☐ Audit Fedora server

bash

```
ssh fedora
sudo lynis audit system

# Note the hardening index: ____%
# Save report
sudo cp /var/log/lynis.log ~/lynis-fedora-$(date +%Y%m%d).log
```

☐ Audit Pi devices

bash

```
# Pi 4
ssh pi4
sudo lynis audit system

# Pi Zero
ssh pi-zero
sudo lynis audit system
```

Create Findings Document

☐ Document vulnerabilities found

bash

```
nano ~/security-findings.md

# Security Findings Report
Date: $(date)

## Nmap Findings

### Fedora Server (100.64.0.10)
- Open Ports:
  - Port 22 (SSH) - Expected ✓
  - Port 3000 (Gitea) - Expected ✓
  - Port 514 (Syslog) - Expected ✓
- Unexpected ports: [List any]

### Pi 4 (100.64.0.30)
- Open Ports: [List]
- Concerns: [List any]

### Pi Zero (100.64.0.20)
- Open Ports: [List]
- Concerns: [List any]

## Lynis Findings

### Critical Issues
1. [Issue description]
  - Severity: High/Medium/Low
  - Remediation: [Steps to fix]

### Warnings
1. [Warning description]
  - Action needed: [Y/N]
  - Fix: [Steps]

### Suggestions
1. [Suggestion]
  - Priority: High/Medium/Low
  - Implement: [Y/N]
```

Common Security Issues & Fixes

☐ **Disable root SSH login** (if not already)

bash

```
sudo nano /etc/ssh/sshd_config
# Set: PermitRootLogin no
sudo systemctl restart sshd
```

☐ Remove unnecessary packages

bash

```
# List installed packages
dpkg -l | less # Ubuntu
rpm -qa | less # Fedora

# Remove what you don't need
sudo apt remove package_name
# OR
sudo dnf remove package_name
```

☐ Check for security updates

bash

```
# Ubuntu:
sudo apt update
sudo apt list --upgradable
sudo apt upgrade

# Fedora:
sudo dnf check-update
sudo dnf upgrade
```

☐ Review running services

bash

```
systemctl list-units --type=service --state=running

# Disable unnecessary services:
sudo systemctl disable SERVICE_NAME
sudo systemctl stop SERVICE_NAME
```

☐ Check file permissions on sensitive files

bash

```
# SSH keys should be 600
ls -l ~/.ssh/id_*
chmod 600 ~/.ssh/id_ed25519

# authorized_keys should be 600
chmod 600 ~/.ssh/authorized_keys

# /etc/shadow should be 000 or 600
ls -l /etc/shadow
```

Create Remediation Plan

☐ **Prioritize fixes**

bash

```
nano ~/remediation-plan.md

# Security Remediation Plan

## Priority 1 - Critical (Fix Immediately)
1. [ ] Disable root SSH on all devices
2. [ ] Update all packages with security patches
3. [ ] Fix any open ports not in use

## Priority 2 - High (Fix This Week)
1. [ ] Remove unnecessary services
2. [ ] Implement fail2ban on all SSH servers
3. [ ] Review and tighten file permissions

## Priority 3 - Medium (Fix This Month)
1. [ ] Install automatic security updates
2. [ ] Implement log monitoring/alerting
3. [ ] Document firewall rules

## Priority 4 - Low (Ongoing)
1. [ ] Regular vulnerability scans (monthly)
2. [ ] Review user accounts quarterly
3. [ ] Keep security documentation updated
```

Optional: Install fail2ban

fail2ban blocks IPs after failed login attempts.

☐ Install fail2ban

bash

```
# Ubuntu:
sudo apt install fail2ban

# Fedora:
sudo dnf install fail2ban
```

☐ Configure fail2ban

bash

```
# Copy default config
sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local

# Edit config
sudo nano /etc/fail2ban/jail.local

# Enable SSH protection:
[sshd]
enabled = true
port = ssh
logpath = %(sshd_log)s
maxretry = 3
bantime = 3600

# Save and restart
sudo systemctl restart fail2ban
sudo systemctl enable fail2ban
```

☐ Check fail2ban status

bash

```
sudo fail2ban-client status
sudo fail2ban-client status sshd
```

1. `nmap`
2. `nmap -sV`
3. `nmap -p`
4. `nmap -A`
5. `lynis`
6. `lynis audit system`
7. `fail2ban`
8. `fail2ban-client`
9. `chkrootkit`
10. `rkhunter`

Concepts:

- What is CVE?
 - What is CVSS?
 - Port scanning ethics?
 - Service fingerprinting?
 - Attack surface?
-

EXAM OBJECTIVES

[https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

NOTES

Scan results summary:

- Fedora Server: _____
- Pi 4: _____
- Pi Zero: _____

Lynis hardening index:

- Laptop: _____%
- Fedora: _____%
- Pi 4: _____%
- Pi Zero: _____%

Critical issues found:

Issues remediated:

Completed: ☐

QUICK REFERENCE

Nmap Cheat Sheet:

```
bash

nmap IP                # Basic scan
nmap -sV IP            # Service version
nmap -p 1-65535 IP     # All ports
nmap -A IP             # Aggressive (OS+version+scripts)
nmap -F IP             # Fast (top 100 ports)
nmap -oN file.txt IP   # Save to file
nmap 192.168.1.0/24    # Scan subnet
```

Common Ports to Check:

- 21: FTP
- 22: SSH
- 23: Telnet (should be closed!)
- 25: SMTP
- 53: DNS
- 80: HTTP
- 443: HTTPS
- 3306: MySQL
- 5432: PostgreSQL

Lynis Quick Commands:

```
bash
```

```
sudo lynis audit system           # Full audit
sudo lynis show settings         # Show config
sudo lynis show version          # Version info
sudo lynis update info           # Update info
```

Security Hardening Checklist:

- ☐ Disable root SSH
- ☐ Key-based SSH only
- ☐ Firewall configured
- ☐ Unnecessary services disabled
- ☐ Regular updates enabled
- ☐ fail2ban installed
- ☐ Strong passwords enforced
- ☐ File permissions correct
- ☐ Log monitoring active
- ☐ Regular vulnerability scans

Previous Week: Package Management (dnf)

Next Week: SELinux & Security Modules